

Student Name: _____	Student ID: _____	Section : C
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Question 1:

A startup is building a smart fitness tracking platform similar to Fitbit integrated with cloud analytics like Google Fit. Below is a simplified system architecture description:

- Users interact with a Mobile App UI to Register/login, Log workouts and View fitness reports
- The Mobile App sends data to a Backend Server via APIs.
- The Backend Server Stores user data in a User Database, Stores workout logs in a Workout Database, Processes and generates fitness reports
- The system integrates with A Wearable Device API (external system providing heart rate & step data), A Payment Gateway for premium subscriptions
- Admins use a Web Admin Panel to Add/remove users, Manage subscriptions, View analytics reports

- 3 a) From the above architecture, identify and classify them as inputs, outputs, queries, ILF, EIF.
 b) Classify each component as an average complexity.
 c) Find functional points
 d) Given the Productivity rate = 10 FP per person-month
 Calculate: Total development effort (in person-months). If the team size is 5 developers, estimate the project duration (in months)

Type	Simple	Average	Complex
EI	4	5	6
EO	5	7	10
EQ	4	5	7
ILF	7	10	15
EIF	5	7	10

a) $EI = 4$ (user reg/login, log workout, manage users (Admin), manage subscription (Admin))
 $EO = 3$ (fitness report, Analytic report, payment receipt)
 $EQ = 2$ (view profile, view work history)
 $EIF = 2$ (Payment gateway, wearable device API)
 $ILF = 3$ (User db, workout db, Subs. db)

b) $UFP = 4 \times 5 + 3 \times 7 + 2 \times 5 + 2 \times 7 + 3 \times 10$

$UFP = 20 + 21 + 10 + 14 + 30 = 95$

$FP = UFP \times CAF$
 $= 95 \times 1.07$

$FP = 101.65$

Let $EF_i = 14 \times 3 = 42$
 $CAF = 0.65 + (0.01 \times EF_i)$
 $0.65 + 0.01 \times 42$
 $CAF = 1.07$

d) $Effort = FP / \text{productivity rate} = 101.65 / 10$
 $Effort = 10.165$
 $duration = Effort / 5 = 2.033$

Solution

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Question 1:

A startup is developing a mental health support platform similar to BetterHelp. The system includes features for users, therapists, and administrators. The following functional components are identified:

1. External Inputs (EI)

- User registration and profile setup (simple) $4 + 4 + 5 + 5 + 6 = 20$
- Booking a therapy session (average)
- Submitting mood journal entries (average)
- Admin adding therapist profiles (complex)

2. External Outputs (EO)

- Session confirmation notifications (simple)
- Monthly progress reports for users (complex)
- Payment receipts (average) $5 + 10 + 7 = 22$

3. External Inquiries (EQ)

- Search therapists by specialization (average)
- View session history (simple) $4 + 5 = 9$

Type	Simple	Average	Complex
EI 4	4	5	6
EO 3	5	7	10
EQ 2	4	5	7
ILF 3	7	10	15
EIF 2	5	7	10

4. Internal Logical Files (ILF)

- User database (complex)
 - Therapist database (average) $15 + 10 + 15 = 40$
 - Session records (complex)
- Standard weight table

5. External Interface Files (EIF)

- Payment gateway system (average)
- Third-party analytics service (simple) $7 + 5 = 12$

Tasks:

1. Calculate the Unadjusted Function Points. $UFP = 20 + 22 + 9 + 40 + 12 = 103$
2. Given the complexity Adjustment Factor = 1.08, compute the Function Points. 111.24
3. If the development team's productivity rate is 12 FP per person-month, estimate the total effort required. $FP/Pr. = 111.24 / 12 = 9.27 \text{ pm}$
4. Briefly explain how Functional Point Analysis differs from LOC-based estimation.

Solution

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Question 1:

A startup is launching a food delivery mobile app similar to Foodpanda. The team wants to plan their release using project scheduling techniques.

The following tasks are identified:

Task ID	Task Description	Dependencies	Duration (days)
T1	Market Research & Idea Validation	—	3
T2	UI/UX Design (Figma Prototypes)	T1	4
T3	Backend API Development	T1	6
T4	Payment Gateway Integration	T3	3
T5	App Frontend Development	T2	5
T6	Real-time Order Tracking Feature	T3	4
T7	System Integration	T4, T5, T6	3
T8	Beta Testing with Users	T7	4
T9	App Deployment (Play Store)	T8	2

Tasks:

- Construct the project network diagram
- Determine the critical path using the Critical Path Method.
- Calculate the minimum time required to launch the app. 22
- Identify which tasks can be delayed without affecting the launch date (i.e., slack/float). T_2, T_4, T_5
- Suppose Real-time Order Tracking (T6) is delayed by 2 days. Analyze its impact on the overall schedule. 24 *last date*

